

Effect of anharmonicity on first-principles calculations of nonradiative carrier capture coefficients [NOTE: modified]

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Determining nonradiative recombination rates at point defect sites is of significant interest both within defect modeling and towards an extensive range of applications. However, despite numerous first-principles approaches to accurately resolve nonradiative recombination behavior via multiphonon electronic transitions, no clear theoretical approach has emerged. We develop a general theory of carrier capture transitions for defect centers whose phonon modes correspond to anharmonic defect-well potentials. Using *ab-initio* density functional theory calculations to characterize defect structures, we evaluate phonon-phonon matrix elements to obtain capture coefficient values at GaN: C_N centers. We determine that effective energy barrier values approach the classical energy barrier upon utilizing an ansatz for the high-temperature dependence of capture coefficients, which verifies the correspondence principle for the anharmonic situation. Moreover, we develop an explicit approximation for low-temperature capture coefficients under anharmonicity.

I. CAPTURE COEFFICIENT CALCULATIONS: FORMALISM

A. First-Principles Derivation

We concentrate our transition-state model on capture and recombination processes that occur by multiphonon emission (MPE). We first offer a general description of the methodology by which we compute capture coefficients; our approach largely synthesizes the approaches of CITE. Following the seminal characterization of Landau and Zener (CITE), we may express the relevant capture cross-section σ in terms of individual transition probabilities (add equation if necessary).

Following Refs. (Alk)/(HL)/(ShiWang), we specify the relevant many-body Hamiltonian for a lattice with a point acceptor defect as

$$\hat{H} = \hat{T}_I + \hat{T}_e + \hat{V}_{II} + \hat{V}_{ee} + \hat{V}_{Ie}, \quad (1)$$

where e and I denote electronic and ionic components, respectively. Upon denoting ionic coordinates by $\{Q\}$ and electronic coordinates by $\{x\}$, as in Ref. (Alk), we express the corresponding many-body wavefunctions as $\Phi(\{Q\}, \{x\})$.

Assuming that the relevant transition probability is within first-order of electron-phonon coupling and that the static approximation holds, we may factor the many-body wavefunctions $\Phi(\{Q\}, \{x\})$ as $\Phi_n(\{Q\}, \{x\}) = \Psi(\{Q\}, \{x\})\phi(Q)$, where $\Psi(\{Q\}, \{x\})$ and $\phi(Q)$ denote electronic and ionic components, respectively. As noted in Ref. (Alk), the capture rate c_n may be determined by means of the following Fermi's golden rule expression:

$$c_n = \frac{2\pi}{\hbar} g \sum_{\text{states } m} w_m \sum_{\text{states } n} |\Delta H_{im;fn}^{\text{e-ph}}|^2 \delta(E_{i,m} - E_{f,n}), \quad (2)$$

where g corresponds to the degeneracy factor of the chosen defect; w_m gives the thermal occupation of state m ; $E_{i,m}, E_{f,n}$ are the energies of the initial and final vibrational states, respectively; and $\Delta H_{im;fn}^{\text{e-ph}}$ is the perturbation in the complete many-body Hamiltonian under a large lattice relaxation. Therefore, upon considering lattice relaxation from lattice coordinate $\{Q\}$ to ground-state center $\{Q_0\}$ and expanding $\Delta H_{im;fn}^{\text{e-ph}}$ under the static approximation, equation (2) may be reformulated as

$$C_p = \frac{2\pi g V}{\hbar} \left\langle \psi_i \left| \frac{\partial \hat{H}}{\partial Q} \right| \psi_f \right\rangle \sum_{\text{states } m} w_m \sum_{\text{states } n} [\langle \phi_{i,m} | Q - Q_0 | \phi_{f,n} \rangle]^2 \delta(E_{i,m} - E_{f,n}) \quad (3)$$

where $\psi_{i,f}$ are the corresponding wavefunctions of the electronic Hamiltonian, V is the relevant supercell volume, and C_0 gives the capture coefficient as opposed to the capture rate. Notably, existing approximations obtain the above formulation under an additional assumption that the multidimensional perturbation in the lattice $\{Q\}$ may be collapsed into a single effective phonon mode (cf. Refs. (Alk)/(ShiWang)). Despite equation (3) being derived in Ref. (Alk), further details are required to explicitly and accurately evaluate the relevant matrix elements.

Moreover, although Ref. (ShiWang) provides several systematic methodologies for evaluating electron-phonon coupling matrix elements, the approach for evaluating phonon-phonon matrix elements assumes harmonicity of the relevant potential energy surfaces; accordingly, it is imperative to develop a model for calculation of such modes in high generality. We describe two such methods in the following subsection.

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B. Phonon-Phonon Matrix Elements

Our goal is to effectively evaluate the inner sum within equation (3) such that the phonon wavefunctions $\phi_{i,m}$, $\phi_{f,n}$ are eigenstates of a general anharmonic potential. Explicitly, for all nonnegative constants α and lattice potentials $V(Q)$, all phonon eigenfunctions $\psi_\alpha(Q)$ must satisfy the time-independent Schrödinger equation under the vibrational Hamiltonian, which gives:

$$\left[\frac{-\hbar^2}{2m_{\text{eff}}} \nabla^2 + \hat{V}(Q) \right] |\phi_\alpha\rangle = \varepsilon_\alpha |\phi_\alpha\rangle, \quad (4)$$

with m_{eff} providing an effective mass under contributions of local lattice atoms. We address this eigenvalue problem using two distinct methods.

1. Mesh-based Approach

We may construct an explicit one-dimensional mesh in configuration coordinate Q , namely $\mathbf{Q} = [q_0, q_1, \dots, q_n]$, on which the values of the eigenfunction ϕ_a may be evaluated. Upon utilization of a one-dimensional coordinate, the kinetic energy operator reduces to $\frac{-\hbar^2}{2m_{\text{eff}}}(\partial_Q)^2$, where ∂_Q denotes a derivative with respect to Q , and may be readily evaluated using a first- or second-order finite difference schema on $\mathbf{Q}$ with respect to mesh spacing $\Delta q := q_1 - q_0$. Furthermore, since the general one-dimensional anharmonic potential $V(Q)$ of degree n is given by

$$V(Q; \kappa_j, \omega) = \kappa_0 + \alpha(\hbar\omega)^2 Q^2 + \sum_{j=3}^n \kappa_j Q^j \quad (5)$$

for coefficients κ_j , the action of the potential energy operator $\hat{V}(Q)$ may be characterized on $\mathbf{Q}$ by the matrix

$$\mathbf{V} = \text{diag}(V(q_0), V(q_1), \dots, V(q_n)).$$

Thus, we may determine $\phi_{i,m}$ and $\phi_{f,n}$ on $\mathbf{Q}$ using Davidson diagonalization on equation (4) with anharmonic potentials $V_i(Q)$, $V_f(Q)$, each of which contain a unique set of coefficients $\{\kappa_j, \omega\}$ that may be evaluated by *ab-initio* calculations (see Sec. REF).

We may also precondition the eigendecomposition procedure through analytic solutions ϕ_m^{harm} , ϕ_n^{harm} to the situation of a harmonic-oscillator potential, in which we may set $\kappa_j = 0$ for $j \geq 3$ in equation (5). As such, we evaluate the relevant phonon-phonon matrix element $\langle \phi_{i,m} | Q - Q_0 | \phi_{f,n} \rangle$, which we denote as O_{ij} , by numerically approximating the overlap integral, as follows:

$$\begin{aligned} O_{m,n} &= \int_{\mathbb{R}} \phi_{i,m}(Q - Q_0)(Q - Q_0)\phi_{f,n}(Q) dQ \\ &\approx \Delta q \sum_{\substack{q \in \mathbf{Q} \\ q_i \geq q_m}} \overline{\phi_{i,m}(q_i - Q_0)}(q_i - Q_0)\phi_{f,n}(q_i), \end{aligned}$$

where $q_m := q_0 + Q_0$ is the minimum value within $\mathbf{Q}$ for which the displaced wavefunction $\phi_{i,m}(Q - Q_0)$ may be evaluated and $\overline{\phi_{i,m}}$ gives a spline interpolation of $\phi_{i,m}(\mathbf{Q})$. Note that the approximation of the explicit integral is motivated by a choice of sufficiently large mesh size $\{Q\}$ such that the residual terms

$$\begin{aligned} r_1 &= \int_{\max(\mathbf{Q})}^{\infty} \phi_{i,m}(Q - Q_0)(Q - Q_0)\phi_{f,n}(Q) dQ, \\ r_2 &= \int_{-\infty}^{q_m} \phi_{i,m}(Q - Q_0)(Q - Q_0)\phi_{f,n}(Q) dQ \end{aligned}$$

are of sufficiently small order to have negligible effects on matrix element values. To be certain that we have adequately bounded the tail integrals, we may approximate the anharmonic solutions by analytic harmonic-oscillator eigenstates at large Q . Applying this for the first residual gives, for instance,

$$\begin{aligned} r_1 &\approx \int_{\max(\mathbf{Q})}^{\infty} \phi_{i,m}^{\text{harm}}(Q - Q_0)(Q - Q_0)\phi_{f,n}^{\text{harm}}(Q) dQ \\ &= \frac{\sqrt{\xi_i \xi_f}}{\sqrt{2^m m! 2^n n! \pi}} \int_{\max(\mathbf{Q})}^{\infty} H_m(\xi_i Q) H_n(\xi_f(Q - Q_0)) \\ &\quad \cdot \exp(-\xi_i^2 Q^2/2 - \xi_f^2(Q - Q_0)^2/2)(Q - Q_0) dQ, \end{aligned}$$

where ξ_i, ξ_f are given by $\sqrt{\frac{\omega_i}{\hbar}}, \sqrt{\frac{\omega_f}{\hbar}}$ respectively and H_m, H_n are Hermite polynomials. Since we may bound Hermite polynomials above by their leading term (i.e., $H_n(x) \leq 2^n x^n$ when $x \geq 0$), we have

$$\begin{aligned} r_1 &\leq \frac{\sqrt{\xi_i \xi_f}}{\sqrt{2^{m+n} m! n! \pi}} \int_{\max(\mathbf{Q})}^{\infty} [2^m (\xi_i Q)^m \exp(-\xi_i^2 Q^2/2)] \cdot (Q - Q_0) \\ &\quad \cdot [2^n (\xi_f(Q - Q_0))^n \exp(-\xi_f^2(Q - Q_0)^2/2)] dQ \\ &\leq \frac{\sqrt{2^{m+n} \xi_i \xi_f}}{\sqrt{m! n! \pi}} \int_{\max(\mathbf{Q})}^{\infty} [(\xi_i Q)^m \exp(-\xi_i^2 Q^2/2)] \\ &\quad \cdot [(\xi_f(Q - Q_0))^{n+1} \exp(-\xi_f^2(Q - Q_0)^2/2)] dQ \\ &= \frac{\sqrt{\xi_i \xi_f 2^{m+n}}}{\sqrt{m! n! \pi}} \int_{\max(\mathbf{Q})}^{\infty} f_m(\xi_i Q) \cdot f_{n+1}(\xi_f(Q - Q_0)) dQ, \end{aligned} \quad (6)$$

where f_α is defined for any state α as $f_\alpha(x) := x^\alpha \exp(-x^2/2)$. Note that we may explicitly evaluate the tail integral of f , since, for $a > 0$,

$$\int_a^\infty x^\alpha \exp(-x^2/2) = 2^{(\alpha-1)/2} \cdot \Gamma\left(\frac{\alpha+1}{2}, \frac{a}{2}\right), \quad (7)$$

where Γ denotes the incomplete Gamma function. Thus, since $f_m(Q)$ and $f_{n+1}(Q - Q_0)$ — with domains $(\max(\mathbf{Q}), \infty)$ and $(\max(\mathbf{Q}) - Q_0, \infty)$ — are nonnegative and integrable, and since $L^2(\max(\mathbf{Q}) - Q_0, \infty)$ is an inner product space, the Cauchy-Schwarz inequality may

be applied to the integrand in (6), which gives:

$$\begin{aligned} & \left[\int_{\max(\mathbf{Q})}^{\infty} f_m(\xi_i Q) \cdot f_{n+1}(\xi_f(Q - Q_0)) dQ \right]^2 \\ & \leq \left[\int_{q_n}^{\infty} f_m(\xi_i Q) dQ \right]^2 \left[\int_{q_n}^{\infty} f_{n+1}(\xi_f(Q - Q_0)) dQ \right]^2 \\ & = 2^{(m-1)/2} \cdot \Gamma\left(\frac{m+1}{2}, \frac{\xi_i q_n}{2}\right) \\ & \quad \cdot 2^{n/2} \cdot \Gamma\left(\frac{n+2}{2}, \frac{\xi_f(q_n - Q_0)}{2}\right). \end{aligned}$$

Applying this bound to (6) gives an explicit bound on the tail integral, namely:

$$\begin{aligned} r_1 & \leq \frac{2^{m+n-1/2} \sqrt{\xi_i \xi_f}}{\sqrt{m!n!}\pi} \Gamma\left(\frac{m+1}{2}, \frac{\xi_i q_n}{2}\right) \\ & \quad \Gamma\left(\frac{n+2}{2}, \frac{\xi_f(q_n - Q_0)}{2}\right). \end{aligned}$$

A similar bound can be developed for the other tail integral, r_2 . Thus, we may set m, n to be the maximum eigenstates in consideration and ensure that q_n is chosen so that the relevant deviations are smaller than $\mathcal{O}(10^{-20})$.

For practical calculations, however, a choice of $\max(\mathbf{Q}) - Q_0 = \min(\mathbf{Q}) = 20$ is suitable to ensure extremely small tail integrals. An alternative method of verifying this calculation is to note that the relevant eigenfunctions decay sharply outside of the classical turning point, which can be evaluated as $Q - Q_0 = \frac{n+1/2}{\alpha\hbar\omega}$, where α, ω are defined as in (5). As such, since $\alpha \approx 119.6$ is constant and $\hbar\omega \geq 0.03$ in typical calculations, a value of $\max(\mathbf{Q}) - Q_0 = 20$ ensures that all states with $\max(m, n) \leq 1430$ decay strongly within the mesh $\mathbf{Q}$, which is certainly sufficient for our desired calculations.

2. Basis-Expansion Approach

To develop a method with greater efficiency and stability than the mesh-based approach, we may represent the relevant anharmonic phonon eigenstates within specially chosen bases — namely, bases of eigenstates of the Hamiltonian corresponding to harmonic potentials.

We assume for simplicity that both potential energy surfaces have equal curvatures (i.e., we have $\hbar\omega := \hbar\omega_i = \hbar\omega_f$); we defer the situation of unequal curvatures to Appendix A. Then, if we assume that the initial state is centered about perturbed lattice coordinate $Q - Q_0$, we may express the general one-dimensional Hamiltonians for both initial and final states as follows:

$$\hat{\mathcal{H}}_{\text{harm}}^i = \frac{\hat{p}^2}{2m} + \frac{1}{2}k(\hat{x} - Q_0)^2, \quad \hat{\mathcal{H}}_{\text{harm}}^f = \frac{\hat{p}^2}{2m} + \frac{1}{2}k\hat{x}^2, \quad (8)$$

for a fixed constant $k = \alpha(\hbar\omega)^2$. The eigenfunctions of both Hamiltonians may be denoted as $\mathcal{B}_1 = \{\psi_{i,m}\}$

and $\mathcal{B}_2 = \{\psi_{f,n}\}$, respectively, where $m, n \geq 0$. Note that both sets of eigenfunctions form bases of the Hilbert space $L^2(\mathbb{R})$. As such, for anharmonic Hamiltonians of the form

$$\begin{aligned} \hat{\mathcal{H}}_{\text{anharm}}^i &= \frac{\hat{p}^2}{2m} + \frac{1}{2}k(\hat{x} - Q_0)^2 + \sum_{j=3}^n \alpha_j (\hat{x} - Q_0)^j, \\ \hat{\mathcal{H}}_{\text{anharm}}^f &= \frac{\hat{p}^2}{2m} + \frac{1}{2}k\hat{x}^2 + \sum_{j=3}^n \beta_j \hat{x}^j, \end{aligned}$$

with suitable coefficients α_j, β_j , we may decompose such Hamiltonians in bases $\mathcal{B}_1, \mathcal{B}_2$, respectively, as

$$\{\hat{\mathcal{H}}_{\text{anharm}}^i\}_{\mathcal{B}_1} = \{\hat{\mathcal{H}}_{\text{harm}}^i\}_{\mathcal{B}_1} + \left\{ \sum_{j=3}^n \alpha_j (\hat{x} - Q_0)^j \right\}_{\mathcal{B}_1}, \quad (9)$$

$$\{\hat{\mathcal{H}}_{\text{anharm}}^f\}_{\mathcal{B}_1} = \{\hat{\mathcal{H}}_{\text{harm}}^f\}_{\mathcal{B}_1} + \left\{ \sum_{j=3}^n \beta_j \hat{x}^j \right\}_{\mathcal{B}_2}. \quad (10)$$

Moreover, since both Hamiltonians are diagonal in the basis given by their eigenstates, we have

$$\begin{aligned} \{\hat{\mathcal{H}}_{\text{anharm}}^i\}_{\mathcal{B}_1} &= \text{diag}\left(\frac{\hbar\omega}{2}, \frac{3\hbar\omega}{2}, \dots\right) + \sum_{j=3}^n \alpha_j \{\hat{x} - Q_0\}_{\mathcal{B}_1}^j, \\ \{\hat{\mathcal{H}}_{\text{anharm}}^f\}_{\mathcal{B}_1} &= \text{diag}\left(\frac{\hbar\omega}{2}, \frac{3\hbar\omega}{2}, \dots\right) + \sum_{j=3}^n \beta_j \{\hat{x}\}_{\mathcal{B}_1}^j. \end{aligned} \quad (11)$$

Finally, we may decompose $\hat{x}$ in $\mathcal{B}_2$ by means of raising and lowering operators:

$$\begin{aligned} \langle \psi_{f,n_1} | \hat{x} | \psi_{f,n_2} \rangle &= \sqrt{\frac{\hbar}{2m\omega}} \langle \psi_{f,n_1} | \hat{a}_+ + \hat{a}_- | \psi_{f,n_2} \rangle \\ &= \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{n_2+1} \delta_{n_1, n_2+1} + \sqrt{n_2} \delta_{n_1, n_2-1}), \end{aligned} \quad (12)$$

which yields identical hollow tridiagonal matrix representations of both $\{\hat{x} - Q_0\}_{\mathcal{B}_1}$ and $\{\hat{x}\}_{\mathcal{B}_2}$. Combining equations (11–12) gives a direct matrix representation of the action of the desired anharmonic Hamiltonian operator in the respective basis.

We will now discuss the calculation of the phonon-phonon overlap element O_{mn} . For any nonnegative m, n , we may decompose the both the initial and final wavefunctions in the bases $\mathcal{B}_1, \mathcal{B}_2$ as follows:

$$\begin{aligned} \langle \phi_{i,m} | &= \sum_{\alpha \in \sigma(\hat{H}^i)} \langle \phi_{i,m} | \psi_{i,\alpha} \rangle \langle \psi_{i,\alpha} |, \\ | \phi_{f,n} \rangle &= \sum_{\beta \in \sigma(\hat{H}^f)} | \psi_{f,\beta} \rangle \langle \psi_{f,\beta} | \phi_{f,n} \rangle, \end{aligned}$$

where $\sigma(\hat{H}^i), \sigma(\hat{H}^f)$ denote the spectra of anharmonic Hamiltonians $\hat{\mathcal{H}}_{\text{anharm}}^i, \hat{\mathcal{H}}_{\text{anharm}}^f$. Substituting the

above representations in the overlap element expression $\langle \phi_{i,m} | Q - Q_0 | \phi_{f,n} \rangle$ gives

$$O_{m,n} = \sum_{\alpha \in \sigma(\hat{H}^i)} \sum_{\beta \in \sigma(\hat{H}^f)} \langle \phi_{i,m} | \psi_{i,\alpha} \rangle \cdot \langle \psi_{i,\alpha} | Q - Q_0 | \psi_{f,\beta} \rangle \langle \psi_{f,\beta} | \phi_{f,n} \rangle.$$

In order to avoid calculating the inner overlap element, which requires an explicit integration procedure with an additional $(Q - Q_0)$ term, we may introduce an additional set of basis states $\psi_{i,\delta}(Q - Q_0) \subset \mathcal{B}_1$, which gives

$$O_{m,n} = \sum_{\alpha \in \sigma(\hat{H}^i)} \sum_{\beta \in \sigma(\hat{H}^f)} \sum_{\delta \in \sigma(\hat{H}^i)} \langle \phi_{i,m} | \psi_{i,\alpha} \rangle \cdot \langle \psi_{i,\alpha} | \hat{x} - Q_0 | \psi_{i,\delta} \rangle \langle \psi_{i,\delta} | \psi_{f,\beta} \rangle \langle \psi_{f,\beta} | \phi_{f,n} \rangle. \quad (13)$$

As such, we may evaluate the complete overlap matrix elements through multiplication of: (i) matrices containing the representations of $\phi_{i,m}, \phi_{f,n}$ in bases $\mathcal{B}_1, \mathcal{B}_2$, which are determined from the band-matrix representations of both anharmonic Hamiltonians; (ii) the matrix containing explicit overlap integrals between displaced harmonic oscillator states, or $\langle \psi_{i,\delta} | \psi_{f,\beta} \rangle$; and (iii) a tridiagonal matrix representation of $\hat{x} - Q_0$ in the basis $\mathcal{B}_1$, which is provided in equation (12).

To determine the explicit overlap integrals $R(\delta, \beta) = \langle \psi_{i,\delta} | \psi_{f,\beta} \rangle$, we must evaluate

$$R(\delta, \beta) = \int_{\mathbb{R}} \psi_{i,\delta}(Q - Q_0) \psi_{f,\beta}(Q) dQ \quad (14)$$

$$= \frac{\sqrt{\alpha}}{\sqrt{\pi} \cdot 2^{m+n} m! n!} \exp\left(-\frac{\alpha(Q^2 + (Q - Q_0)^2)}{2}\right) \cdot H_{\delta}(Q - Q_0) H_{\beta}(Q), \quad (15)$$

with $\alpha := \frac{m\omega}{\hbar} = \xi^2$. Utilizing the explicit formula given by Smith (Ref.) and a simplification of the formula given by Hutchinsonson (Ref.), we obtain

$$R(\delta, \beta) = \sqrt{2^{-m-n} m! n!} (-1)^n \exp(-\gamma^2/4) \cdot \sum_{r=0}^{\min(m,n)} \frac{(-2)^r \gamma^{m+n-2r}}{r!(n-r)!(m-r)!}, \quad (16)$$

where $\gamma = Q_0 \sqrt{\alpha}$. Hence, combining equations (12–16) with a QR diagonalization procedure to obtain representations of anharmonic eigenfunctions in $\mathcal{B}_1, \mathcal{B}_2$, we may readily obtain the overlap elements $O_{m,n}$ over all desired states (m, n) .

We note that, for values of m, n of $\mathcal{O}(10^2)$ or greater, the inner terms of the sum in equation (16) are of extremely high order. However, for sufficient floating-point precision, calculations of $R(\delta, \beta)$ for states with $m, n \geq 500$ exhibit consistency with expensive numerical approximations to $R(\delta, \beta)$.

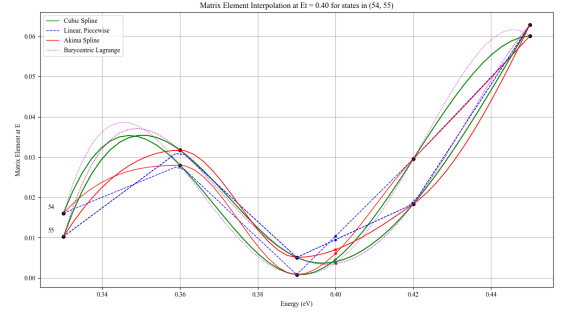


FIG. 1. Underestimation of the interpolated matrix element value upon usage of cubic spline interpolation for a specific eigenstate, while the Akima interpolation method yields a value that lies in the range spanned the two available points that most closely bracket energy E_t .

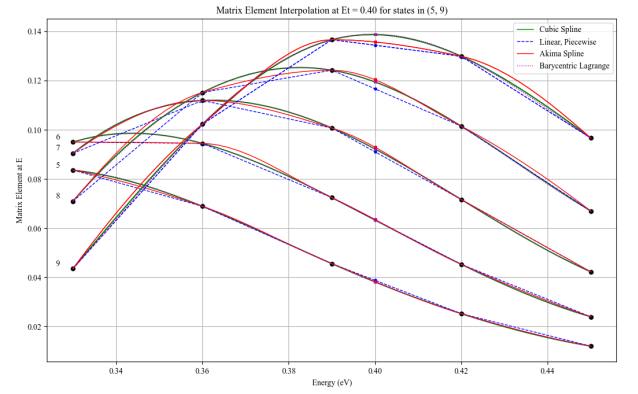


FIG. 2. Overestimation, sim to fig 2 (add caption)

II. NUMERICAL SIMULATIONS

A. Asymptotic Energy Barrier Approximation

In their seminal semiclassical derivation of nonradiative carrier capture, Henry and Lang (Ref.) posited a description of the high-temperature dependence of the capture cross-section σ_c , which may be extended to the capture coefficient C_p as:

$$C_p \sim \frac{C_1}{\sqrt{T}} \cdot \exp\left(-\frac{E_B}{kT}\right), \quad (17)$$

which implies that the classical thermal energy barrier E_B is the key determiner of the value of $C_p(T)$ in the high-temperature limit. However, a systematic verification of this observation utilizing quantum-scale numerical approaches has yet to be performed; as such, we verify the validity of equation (17) across possible values of $\hbar\omega$, trap depth E_t , and lattice displacement Q_0 .

We proceed as follows: for a specific choice of such parameters, we evaluate the classical energy barrier E_B by deriving the energy displacement of the crossing point of potential energy surfaces to the excited-state minimum

$\langle a, b \rangle$, as in Figure **MAKE ONE**, which may readily be evaluated as:

$$E_B = \frac{(E_t - \alpha(\hbar\omega)^2 Q_0^2)^2}{4\alpha(\hbar\omega)^2 Q_0^2}.$$

Moreover, upon computing capture coefficients through the processes described by equations (3) and (13), we fit the resulting high-temperature values of $C_p(T)$ to the ansatz provided by equation (17), which gives a discrete sequence of numerical energy barrier values, denoted $E_B^f(T)$, that ought to converge monotonically to the classical barrier E_B .

In order to specify the tail behavior of $E_B^f(T)$ and overall possible deviations from the predicted E_B , we derive an approximation for the asymptotic behavior of the first finite difference of $E_B^f(T)$ when T becomes large. Across a broad range (**which range**) of parameters ($E_t, Q_0, \hbar\omega_i, \hbar\omega_f$), we obtain the asymptotic approximation

$$\log(E_B^f(T + \Delta T) - E_B^f(T)) \sim \alpha + \beta T,$$

where $\alpha, \beta < 0$ and ΔT is chosen with respect to the particular discrete distribution of temperatures considered. Thus, the end-behavior of the fit energy barrier sequence after evaluation at a threshold temperature a may be effectively determined as follows:

$$\begin{aligned} \lim_{T \rightarrow \infty} E_B^f(T) &= E_B^f(a) + \sum_{T=a}^{\infty} \frac{(E_B^f(T + \Delta T) - E_B^f(T))}{\Delta T} \\ &\leq E_B^f(a) + \frac{1}{\Delta T} \int_a^{\infty} \exp(\alpha + \beta T) dT, \end{aligned}$$

where $\beta < 0$ implies convergence. In Table 1, we display the expected tail sums as well as high-temperature values of $E_B^f(T)$; overall, we observe convergence within 2-3 meV of the expected value of E_B , a significant improvement on the 0.5 eV errors reported by Alkauskas (Ref.) (**Add any extra sources?**). Notably, the residual error between $E_B^f(T)$ and E_B may be resolved under different parameter choices using differing interpolation methods, yet, for implementation purposes, any of the two approaches outlined in **REFSEC** will suffice.

B. High-Temperature Effects of Anharmonicity

$$C_P(T) \propto \frac{1}{\sqrt{T}} \exp\left(-\frac{E_B}{k_B T}\right),$$

Add:

1. Description of the high-temperature anharmonicity from today morning (goes up a little bit more than we thought, and converges well)

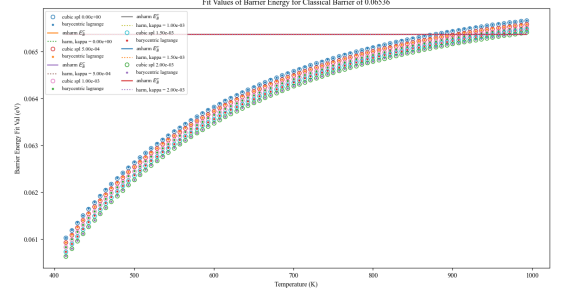


FIG. 3. Values of fit energy barrier $E_B^f(T)$ at various cutoff temperatures T , with increasing anharmonic coefficient κ in $[0, 2 \cdot 10^{-3}]$. The relevant values of defect-well parameters $E_t, Q_0, \hbar\omega_i, \hbar\omega_f$ are (0.4, 1.3, 0.03, 0.03), yielding a classical energy barrier of 0.066 eV; this barrier is reproduced with high accuracy by all choices of anharmonicity.

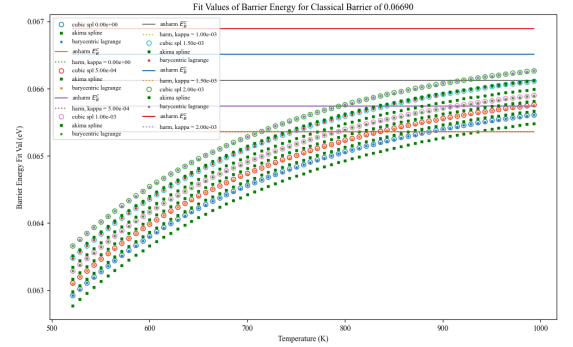


FIG. 4. Values of fit energy barrier $E_B^f(T)$ with increasing anharmonic coefficient κ in $[0, 2 \cdot 10^{-3}]$, where $E_t, Q_0, \hbar\omega_i, \hbar\omega_f = (0.4, 1.3, 0.03, 0.03)$ remain invariant — causing the anharmonic classical barrier to increase in κ . Monotonicity of anharmonic values of $E_B^f(T)$ with respect to κ is observed; higher energy barrier values, moreover, slow the convergence of both the harmonic and anharmonic situations.

2. Discussion of how the $Q - Q_0$ factor's inclusion actually changes whether the results might work well or not.
3. Indication of where the approximation breaks down — when the parabola starts curving downwards because of the necessity of using so many states due to the scale fact at temperatures ≥ 5000 K or so.
4. A different figure about the case when we fix E_B .
5. Ensure to discuss convergence of the numerical solver vs the basis-based solver

Appendix A: Calculation of Overlap Elements for Unequal Curvatures

We extend the results of section (IB 2) by explicitly determining an explicit description of the phonon-phonon overlap integral $O_{m,n} = \langle \phi_{i,m} | Q - Q_0 | \phi_{f,n} \rangle$ to a situation where the relevant potential energy surfaces do not necessarily have equal curvature.

In the situation of a harmonic potential energy distribution, the relevant one-dimensional Hamiltonian operators for initial and final states may be described in the following analogue of equation (8):

$$\hat{\mathcal{H}}_{\text{harm}}^i = \frac{\hat{p}^2}{2m} + \frac{1}{2}k_1(\hat{x} - Q_0)^2, \quad \hat{\mathcal{H}}_{\text{harm}}^f = \frac{\hat{p}^2}{2m} + \frac{1}{2}k_2\hat{x}^2,$$

where $k_1 = \alpha(\hbar\omega_i)^2$, $k_2 = \alpha(\hbar\omega_f)^2$ are the respective curvatures of the potential energy surfaces. Since equations (9-10) hold identically, the analogue of equation (11) for the basis expansions of anharmonic Hamiltonians is clear:

$$\{\hat{\mathcal{H}}_{\text{anharm}}^i\}_{\mathcal{B}_1} = \text{diag}\left(\frac{\hbar\omega_i}{2}, \frac{3\hbar\omega_i}{2}, \dots\right) + \sum_{j=3}^n \alpha_j \{\hat{x} - Q_0\}_{\mathcal{B}_1}^j, \quad (\text{A1})$$

$$\{\hat{\mathcal{H}}_{\text{anharm}}^f\}_{\mathcal{B}_1} = \text{diag}\left(\frac{\hbar\omega_f}{2}, \frac{3\hbar\omega_f}{2}, \dots\right) + \sum_{j=3}^n \beta_j \{\hat{x}\}_{\mathcal{B}_1}^j. \quad (\text{A2})$$

The direct analogue of equation (12) also holds, where ω is replaced by ω_i, ω_f respectively. Thus, the only remaining component to evaluate $O_{m,n}$ using equation (13) is evaluating the overlap integrals $R(\delta, \beta) \langle \psi_{i,\delta} | \psi_{f,\beta} \rangle$ — which appear in (14) — for eigenfunctions that result from potentials of unequal curvatures. To do so, we may note the following result of Engleman for specific characterizations of $R(\delta, \beta)$ when one of δ or β is zero:

$$R(\delta, 0) = \sqrt{2^{-\delta} \cdot qm!} \cdot \exp\left(-\frac{\gamma^2 p}{4}\right) \sum_{t=0}^{\chi(\delta)} \frac{(-x)^t (\gamma q)^{\delta-2t}}{t! (\delta-2t)!}, \quad (\text{A3})$$

$$R(0, \beta) = (-1)^\beta \sqrt{2^{-\beta} \cdot qm!} \cdot \exp\left(-\frac{\gamma^2 p}{4}\right) \sum_{t=0}^{\chi(m)} \frac{x^t (\gamma p)^{\beta-2t}}{t! (\beta-2t)!}, \quad (\text{A4})$$

where the relevant nondimensional parameters (p, q, x) are given by

$$p = \frac{2\omega_i}{2\omega_i\omega_f}, \quad q = \frac{2\sqrt{\omega_i\omega_f}}{\omega_i + \omega_f}, \quad x = \frac{\omega_2 - \omega_1}{\omega_2 + \omega_1};$$

the parameter γ is defined as $Q_0\sqrt{a}$ as in equation (16); and the function $\chi(\tau)$ is given by $\chi(\tau) = \frac{m-\tau'}{2}$, with τ' giving the reduction of τ modulo 2. We may also utilize the recursion relationship for $R(\delta, \beta)$ derived by Smith

(Ref.), which gives

$$R(\delta, \beta) = \sum_{r=0}^{\min(m,n)} q^r \sqrt{\binom{\delta}{r} \binom{\beta}{r}} \frac{R(\delta-r, 0) R(0, \beta-r)}{R(0, 0)}. \quad (\text{A5})$$

Combining equations (A3–A5), we obtain a method of calculating $R(\delta, \beta)$ for arbitrary choice of parameters δ, β and curvatures ω_i, ω_f . Although this method requires greater complexity than the implementation of equation (16), calculation of overlap integrals nonetheless occurs with substantially greater efficiency and accuracy than the mesh-based approach.

Appendix B: Multidimensional Situation

We wish to extend the results established in section IB for potential energy surfaces and wavefunctions defined over two spatial dimensions. To our knowledge, this situation has yet to be described within the relevant literature; as such, we provide a thorough exposition of the approach towards multidimensionality.

The mesh-based generalization is fairly straightforward. We define the mesh $\mathbf{Q} = \mathbf{Q}_1 \times \mathbf{Q}_2$, where $\mathbf{Q}_1, \mathbf{Q}_2$ are both one-dimensional meshes of the form $[x_1, \dots, x_m]$ and $[y_1, \dots, y_n]$, with spacing $\Delta Q_1, \Delta Q_2$. In the harmonic situation, we may specify the potential energy operator as $\hat{V}(x) = \frac{1}{2}k_x x^2 + \frac{1}{2}k_y y^2$; we do not assume isotropy for generality. Thus, we may encapsulate the action of the Hamiltonian operator on a wavefunction $\psi \in L^2(\mathbb{R}^2)$ as

$$\hat{H}|\psi\rangle = \frac{\hbar^2}{2m} \nabla^2 |\psi\rangle + \frac{1}{2} (k_x \hat{x}^2 + k_y \hat{y}^2) |\psi\rangle. \quad (\text{B1})$$

Moreover, we explicitly construct the matrix Ψ of values of ψ on $\mathbf{Q}$ as

$$\Psi = \begin{bmatrix} \psi(x_1, y_1) & \dots & \psi(x_1, y_n) \\ \vdots & \ddots & \vdots \\ \psi(x_m, y_1) & \dots & \psi(x_m, y_n) \end{bmatrix}.$$

If we utilize a first-order finite difference approximation in estimating ψ_{xx}, ψ_{yy} for purposes of the Laplacian operator, we may effectively encapsulate the approximate action of the second derivative operator by

$$D_M = \begin{bmatrix} -2 & 1 & 0 & \dots & 0 \\ 1 & -2 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & -2 & 1 \\ \vdots & \dots & 0 & 1 & -2 \end{bmatrix},$$

so that the effective action of the kinetic energy operator on Ψ may be encapsulated as

$$\hat{T}\Psi = \frac{\hbar^2}{2m} \nabla^2 \Psi = \frac{-\hbar^2}{2m} \left(\frac{D_M \Psi}{\Delta Q_1^2} + \frac{(D_M \Psi)^\dagger}{\Delta Q_2^2} \right). \quad (\text{B2})$$

Moreover, by defining the diagonal matrices

$$V_x = \text{diag} \left(\frac{1}{2} k_x x_1^2, \dots, \frac{1}{2} k_x x_m^2 \right),$$

$$V_y = \text{diag} \left(\frac{1}{2} k_y y_1^2, \dots, \frac{1}{2} k_y y_n^2 \right),$$

we may encapsulate the action of the potential energy

operator on Ψ as $V_x \Psi + (V_y \Psi^\dagger)^\dagger$; combining this expression with equations (B1-B2) gives a complete description of the action of the Hamiltonian:

$$\hat{H} \Psi = \frac{-\hbar^2}{2m} \left(\frac{D_M \Psi}{\Delta Q_1^2} + \frac{(D_M \Psi^\dagger)^\dagger}{\Delta Q_2^2} \right) + V_x \Psi + (V_y \Psi^\dagger)^\dagger.$$